

Sec. IV Role-Aware Operational Legitimacy
Verification across Distributed Nodes

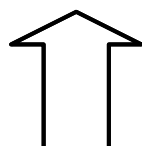
Destination-Side: Output-Based
Heterogenous Execution Verification

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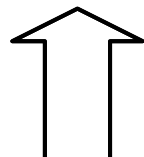
In-Path: Lightweight Admissibility
Enforcement

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Source-Side: Forward Semantic Error
Detection



Security-State
Inferability



Operational
Verifiability

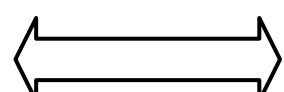
Sec. II Trustworthy Cryptographic Processing under Complex Security-State Management

Minimal Trusted
Crypto Unit

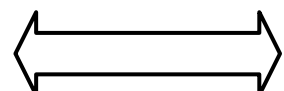
Protected Key
Material Storage

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Cryptographic
Algorithm Execution



Narrow Control
Interface



Per-Message Data
Interface

Operational Security-State Management and
Decision Unit

SA and Key
Lifecycle State

Per-SA Anti-Replay
State

Per-Flow Security-
Mode State



Operational
Security
Decision